

Assignment 5: CASAS Smart-Home Activity Dataset

DA331 Big Data Analytics: Tools & Techniques

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1. Dataset and tools

I used the Kaggle mirror of the CASAS Smart Home dataset (Zenodo 15708568), home **hh101**, which is a single resident with six room-level ambient sensors (PIR motion ON/OFF and door OPEN/CLOSE). It contains two files:

- a **raw stream** of 1,286,244 unlabelled events (2012-07-18 to 2013-07-25);
- a **labelled subset** of 222,858 events (2012-07-20 to 2012-09-17), where activities are marked as Activity="begin" ... Activity="end".

The work was done in Jupyter/Kaggle notebooks with **pandas** (vectorised parsing, categorical dtypes: the one-year stream takes 13 MB in memory), **matplotlib/seaborn** for visualisation and **scikit-learn** for modelling. The begin/end annotations were expanded to one label per event (events outside any segment are labelled *Other*). Each annotated segment was then collapsed into an **activity episode**, giving 2,677 episodes over 34 activity labels.

2. Exploratory data analysis (Task 1, 01_EDA_CASAS.ipynb)

- **Quality:** there are no missing values and no duplicate rows. The labelled file has one out-of-order timestamp, which was fixed by sorting. The only outage is a two-day gap in July 2013. The home averages about 3,500 events per day.
- **Sensors:** motion events make up 99.4% of the data. The living room (33%), bedroom (23%) and kitchen (22%) dominate, and a typical motion trigger lasts 1–2 s. Inter-event times are heavy-tailed (median 1.2 s, 99th percentile 341 s), so the stream is bursty.
- **Circadian rhythm:** activity peaks at 09–12h, and only 8.6% of events occur between 00:00 and 06:00. Kitchen activity peaks at 08–10h, and weekends have about 10% more events than weekdays.
- **Activities:** 27.9% of labelled-file events are *Other*. The labels are highly imbalanced: five activities (Toilet, Watch_TV, Sleep_Out_Of_Bed, Personal_Hygiene, Dress) make up 46% of episodes, and nine labels have fewer than 20 episodes. Durations range from seconds (Wake_Up) to about 5 h (Sleep). Each activity has a distinct time and room signature, e.g. Morning_Meds at 06–08h, Cook/Eat_Dinner at 17–19h and Bed_Toilet_Transition at 02–05h.
- **Transitions:** many transitions are almost deterministic (Go_To_Sleep→Sleep 1.00, Leave_Home→Enter_Home 0.99, Bathe→Dress 0.95). The entropy of the next activity falls from 4.42 bits to 2.64 bits given the current activity, and to 2.02 / 1.79 bits when the time of day / the previous activity is also known. This directly motivates the model design.

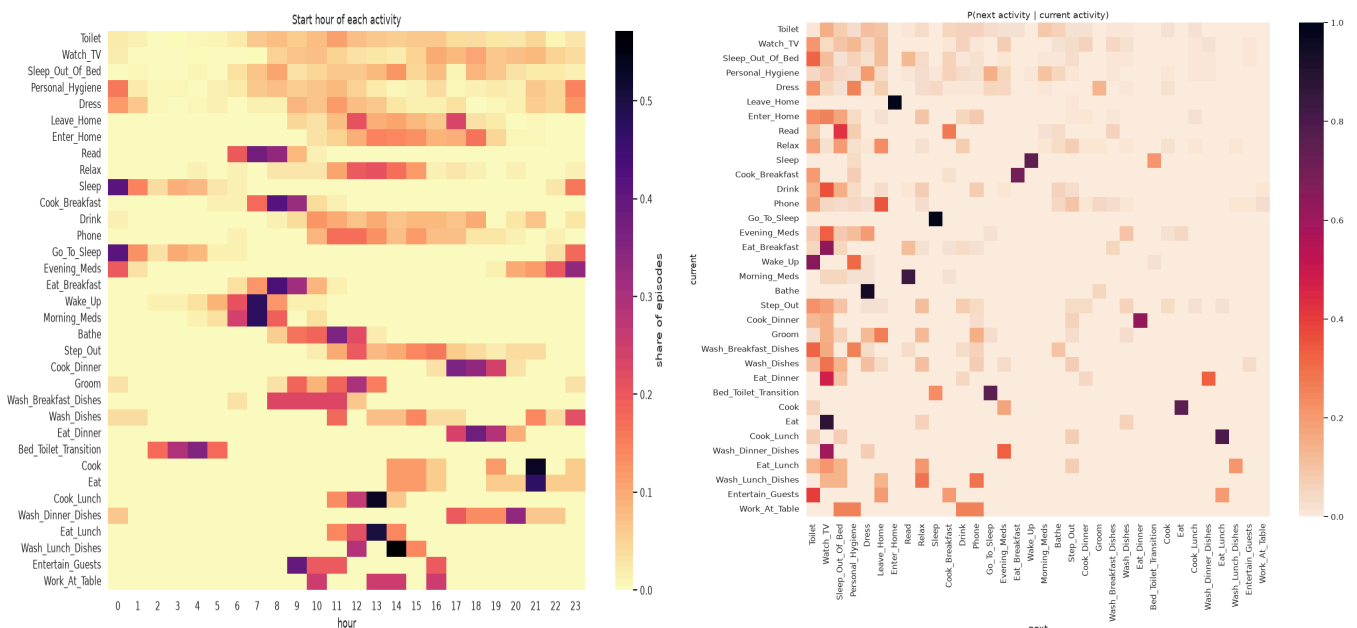


Figure 1. Start-hour profile of each activity (left) and activity transition matrix (right).

3. Next-activity temporal model (Task 2, 02_Next_Activity_Model.ipynb)

Formulation. When episode t ends, predict the activity of episode $t+1$. Nine labels with fewer than 20 episodes are merged into a *Rare* class, which leaves 26 classes. For each episode the features are:

- the current activity and the three previous activities (lags);
- the end time as cyclic $\sin/\cos(\text{hour})$, plus day of week and a weekend flag;
- log duration and log event count of the current episode;
- the last sensor (room) that fired;
- the idle time since the previous episode.

All features are available at prediction time. The data is split **chronologically**: 70% train (Jul 20–Sep 1), 15% validation (Sep 1–9) and 15% test (Sep 9–17, 402 transitions).

Models.

- **Baselines:** majority class, and Markov chains $P(\text{next} \mid \text{current})$, $P(\text{next} \mid \text{previous}, \text{current})$ and the time-aware $P(\text{next} \mid \text{current}, 3\text{-hour bin})$, each with add- α smoothing and back-off.
- **Main model: HistGradientBoostingClassifier** with native categorical features. Three configurations were compared on validation, and the best was refit on train+validation.

Model (test period)	Accuracy	Macro-F1	Top-3 acc.
Majority class	0.102	0.007	0.259
Markov-1: $P(\text{next} \mid \text{cur})$	0.343	0.263	0.609
Markov-2: $P(\text{next} \mid \text{prev}, \text{cur})$	0.415	0.400	0.644
Time-aware Markov: $P(\text{next} \mid \text{cur}, 3\text{h bin})$	0.428	0.385	0.659
Gradient boosting (temporal features)	0.460	0.465	0.692

Analysis. Each source of temporal information improves on the one before: sequence history (Markov-2), time of day (time-aware Markov), and their combination with episode context (gradient boosting). Gradient boosting raises accuracy from 0.34 to 0.46 and macro-F1 from 0.26 to 0.47 over the first-order chain.

- **Feature importance:** by permutation importance, the most useful features are the current activity and the time of day, followed by earlier lags.
- **What works:** routine transitions are predicted well (Enter_Home F1 1.00; Read, Eat_Breakfast, Sleep and Wake_Up F1 ≥ 0.82), and accuracy is highest in the structured morning (72% at 06–09h).
- **What fails:** most errors involve short "filler" activities that can follow almost anything (Toilet, Phone, Drink, Personal_Hygiene) and the heterogeneous *Rare* class.

4. Conclusion

The pipeline runs end to end: raw sensor text, then label expansion, episodes, leakage-free temporal features, and chronological evaluation. The resulting model predicts the next activity with 46% top-1 and 69% top-3 accuracy among 26 classes. Its limitations are a single home and only 60 labelled days. Possible extensions are sequence models (GRU/Transformer) over episodes, semi-supervised use of the 10 unlabelled months, and predicting *when* the next activity will start.